Solutions Tutorial 0

1. Calculate the following limits:

(a) $\lim\limits_{x\to 0}\dfrac{1+x+x^2}{e^x+3}$ (b) $\lim\limits_{x\to 3}\dfrac{x^4-9}{x^2-3}$ (c) $\lim\limits_{x\to\sqrt{3}}\dfrac{x^4-9}{x^2-3}$

Solution.

(a) $\lim\limits_{x\to 0}\dfrac{1+x+x^2}{e^x+3}=\dfrac{\lim\limits_{x\to 0}\left(1+x+x^2\right)}{\lim\limits_{x\to 0}\left(e^x+3\right)}=\dfrac{1+0+0^2}{e^0+3}=\dfrac{1}{4}$.

(b) $\lim\limits_{x\to 3}\dfrac{x^4-9}{x^2-3}=\lim\limits_{x\to 3}\dfrac{(x^2-3)(x^2+3)}{x^2-3}=\lim\limits_{x\to 3}\dfrac{\cancel{(x^2-3)}(x^2+3)}{\cancel{x^2-3}}=\lim\limits_{x\to 3}(x^2+3)=12$.

(c) $\lim\limits_{x\to\sqrt{3}}\dfrac{x^4-9}{x^2-3}=\lim\limits_{x\to\sqrt{3}}(x^2+3)=6$.

2. Given $x_n=\frac{2}{n}$. Calculate $\lim\limits_{n\to\infty}f(x_n)$ for the following functions:

(a) $f(x)=\dfrac{1+x^{-1}}{x^{-2}+2}$ (b) $f(x)=e^x-1$

Solution.

(a) $\lim\limits_{n\to\infty}f(x_n)=\lim\limits_{n\to\infty}\dfrac{1+x_n^{-1}}{x_n^{-2}+2}=\lim\limits_{n\to\infty}\dfrac{1+\frac{n}{2}}{\frac{n^2}{4}+2}=\lim\limits_{n\to\infty}\dfrac{\frac{1}{n^2}+\frac{1}{2n}}{\frac{1}{4}+\frac{2}{n^2}}=0$.

(b) $\lim\limits_{n\to\infty}f(x_n)=\lim\limits_{n\to\infty}(e^{x_n}-1)=\lim\limits_{n\to\infty}(e^{2/n}-1)=e^0-1=0$.

Alternative: $\left(x_n\right)=\left(\frac{2}{n}\right)\to 0$ as $n\to\infty$, so $\lim\limits_{n\to\infty}f(x_n)=\lim\limits_{x\to 0}f(x)$.

(a) $\lim\limits_{n\to\infty}f(x_n)=\lim\limits_{x\to 0}\dfrac{1+\frac{1}{x}}{\frac{1}{x^2}+2}=\lim\limits_{x\to 0}\dfrac{\left(\frac{x+1}{x}\right)}{\left(\frac{1+2x^2}{x^2}\right)}=\lim\limits_{x\to 0}\dfrac{x(x+1)}{1+2x^2}=0$

(b) $\lim\limits_{n\to\infty}f(x_n)=\lim\limits_{x\to 0}(e^x-1)=e^0-1=0$

3. Use that $\lim\limits_{x\to 0}\dfrac{\sin x}{x}=1$ to establish the following limits.

(a) $\lim\limits_{x\to 0}\dfrac{\sin 2x}{x}$ (b) $\lim\limits_{x\to 0}\dfrac{\tan 2x}{\sin x}$ (c) $\lim\limits_{x\to a}\dfrac{\sin x-\sin a}{x-a}$

(d) $\lim\limits_{x\to 0}\dfrac{\sin x-\sin a}{x-a}$ (e) $\lim\limits_{x\to 0}\dfrac{x^2+2x}{\sin x}$ (f) $\lim\limits_{x\to 0}\dfrac{1-\cos x}{x^2}$

Solution.

(a) Let $u=2x$. Then $\lim\limits_{x\to 0}\dfrac{\sin 2x}{x}=2\lim\limits_{x\to 0}\dfrac{\sin 2x}{2x}=2\lim\limits_{u\to 0}\dfrac{\sin u}{u}=2\times 1=2$.

(b) $\lim\limits_{x\to 0}\dfrac{\tan 2x}{\sin x}=\lim\limits_{x\to 0}\dfrac{\sin 2x}{\cos 2x\sin x}=2\lim\limits_{x\to 0}\left(\dfrac{\sin 2x}{2x}\times\dfrac{1}{\cos 2x}\times\dfrac{x}{\sin x}\right)=2\times 1\times 1\times 1=2$.

(c) Let $u=x-a$. Then $x\to a\Rightarrow u\to 0$. Therefore,

$$\begin{aligned}\lim_{x\to a}\frac{\sin x-\sin a}{x-a}&=\lim_{u\to 0}\frac{\sin(u+a)-\sin a}{u}\\&=\lim_{u\to 0}\frac{\sin u\cos a+\cos u\sin a-\sin a}{u}\\&=\lim_{u\to 0}\frac{\sin u\cos a}{u}+\lim_{u\to 0}\frac{\cos u\sin a-\sin a}{u}\\&=\cos a+\sin a\times\lim_{u\to 0}\frac{\cos u-1}{u}\\&=\cos a\end{aligned}$$

Remark: Note that this is the definition of derivative.

(d) $\lim\limits_{x\to 0}\dfrac{\sin x-\sin a}{x-a}=\dfrac{\sin a}{a}$ if $a\neq 0$.

If $a=0$, then $\lim\limits_{x\to 0}\dfrac{\sin x-\sin a}{x-a}=\lim\limits_{x\to 0}\dfrac{\sin x}{x}=1$.

$$\begin{aligned}\text{(e)}\ \lim_{x\to 0}\frac{x^2+2x}{\sin x}&=\lim_{x\to 0}\frac{x^2}{\sin x}+\lim_{x\to 0}\frac{2x}{\sin x}\\&=\left(\lim_{x\to 0}x\right)\cdot\left(\lim_{x\to 0}\frac{x}{\sin x}\right)+2\cdot\lim_{x\to 0}\frac{x}{\sin x}\\&=0(1)+2(1)\\&=2\end{aligned}$$

$$\begin{aligned}\text{(f)}\ \lim_{x\to 0}\frac{1-\cos x}{x^2}&=\lim_{x\to 0}\frac{1-(1-2\sin^2\frac{x}{2})}{x^2}=\lim_{x\to 0}\frac{2(\sin\frac{x}{2})^2}{x^2}\\&=\frac{1}{2}\lim_{x\to 0}\left(\frac{\sin\frac{x}{2}}{\frac{x}{2}}\right)^2\\&=\frac{1}{2}\times 1^2\\&=\frac{1}{2}\end{aligned}$$

4. Find the limits of a_n when $n \to \infty$.

(a) $a_n = \dfrac{1}{n} + \dfrac{6}{n^3}$ (b) $a_n = \dfrac{n^2 - 5}{n^2 + 6n}$

(c) $a_n = \dfrac{5n^2 + 3n - 2}{2n^2 + 7n + 3}$ (d) $a_n = \dfrac{-2n^3 + 4n^2 + 1}{7n^3 + 6n^2 + 5n + 4}$

(e) $a_n = \dfrac{n^5 + n^3 + n}{n^7 + n^5 + n^3 + n}$

Solution. (a) $\lim\left(\dfrac{1}{n} + \dfrac{6}{n^3}\right) = \lim\left(\dfrac{1}{n}\right) + 6 \times \lim\left(\dfrac{6}{n^3}\right) = 0 + 6(0) = 0$

(b) $\lim\left(\dfrac{n^2 - 5}{n^2 + 6n}\right) = \lim\left(\dfrac{1 - \frac{5}{n^2}}{1 + \frac{6}{n}}\right) = \dfrac{1 - 0}{1 + 0} = 1$

(c) $\lim\left(\dfrac{5n^2 + 3n - 2}{2n^2 + 7n + 3}\right) = \lim\left(\dfrac{5 + \frac{3}{n} - \frac{2}{n^2}}{2 + \frac{7}{n} + \frac{3}{n^2}}\right) = \dfrac{5}{2}$

(d) $\lim\left(\dfrac{-2n^3 + 4n^2 + 1}{7n^3 + 6n^2 + 5n + 4}\right) = \lim\left(\dfrac{-2 + \frac{4}{n} + \frac{1}{n^3}}{7 + \frac{6}{n} + \frac{5}{n^2} + \frac{4}{n^3}}\right) = -\dfrac{2}{7}$

(e) $\lim\left(\dfrac{n^5 + n^3 + n}{n^7 + n^5 + n^3 + n}\right) = \lim\left(\dfrac{\frac{1}{n^2} + \frac{1}{n^4} + \frac{1}{n^6}}{1 + \frac{1}{n^2} + \frac{1}{n^4} + \frac{1}{n^6}}\right) = 0$